

Hospital Patient Analytics Report

55,000 Patients . 2019 – 2024

Prepared by: Oluwatobi Alawode, PhD.

Population Health & Health Care Data Analyst

AT A GLANCE — KEY METRICS

Metric	Value	Notes
Total Patient Encounters	55,500	2019–2024, 5-year dataset
Total Revenue	\$1.42 Billion	Across all insurance providers
Average Billing per Encounter	\$25,540	Near-uniform across conditions and payers
Average Length of Stay	15.5 days	Consistent across all admission types
Emergency Admission Rate	33.5%	Similar across all medical conditions
Abnormal Test Result Rate	33.6%	Uniform across conditions and admission types
Unique Insurance Providers	5	Cigna, Medicare, UHC, Blue Cross, Aetna
Medical Conditions Analyzed	6	Arthritis, Diabetes, Hypertension, Obesity, Cancer, Asthma

TABLE OF CONTENTS

Financial & Revenue Questions

Q1. Which insurers account for the highest total billing?	3
Q2. What is the average billing by insurance provider?	3
Q3. Do emergency admissions cost more than elective or urgent?	4
Q4. Which conditions are associated with the highest billing?	4

Utilization & Operations Questions

Q5. What is the average length of stay by admission type?	5
Q6. Are older patients associated with longer stays or higher costs?	5
Q7. Which hospitals handle the highest volume of emergency admissions?	6

Quality & Outcomes Questions

Q8. Are abnormal test results associated with higher billing or longer stays?	7
Q9. Does admission type influence test outcomes?	7
Q10. Are certain conditions more likely to result in emergency admissions?	8

Summary of Findings & Strategic Recommendations	9
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SECTION 1: FINANCIAL & REVENUE ANALYSIS

Q1 Which insurance providers account for the highest total billing?

Domain: Financial & Revenue

This question examines the total revenue contribution of each insurance provider in the dataset. Understanding payer mix and revenue concentration is a core capability in hospital financial analytics — it informs contract negotiation, risk diversification, and revenue cycle strategy.

The analysis aggregated total billing amounts and encounter counts by insurance provider across all 55,500 patient records (2019–2024).

Insurance Provider	Total Revenue	Encounters	Avg per Encounter
Cigna	\$287.2M	11,249	\$25,527
Medicare	\$285.7M	11,154	\$25,617
Blue Cross	\$283.3M	11,059	\$25,614
UnitedHealthcare	\$282.5M	11,125	\$25,390
Aetna	\$278.9M	10,913	\$25,555

KEY FINDING: Cigna leads with \$287.2M in total revenue across 11,249 encounters. All five payers fall within an \$8.3M range - a remarkably balanced payer mix. No single insurer dominates revenue, reducing dependency risk for the health system.

Q2 What is the average billing amount by insurance provider?

Domain: Financial & Revenue

While total revenue reflects volume, average billing per encounter measures the intensity of resource use per patient, independent of patient volume. This metric is central to payer contract benchmarking and case mix analysis.

KEY FINDING: Medicare has the highest average billing at \$25,617 per encounter; UnitedHealthcare the lowest at \$25,390. The total spread of \$227 across all five payers is statistically negligible, indicating uniform resource utilization regardless of insurance type.

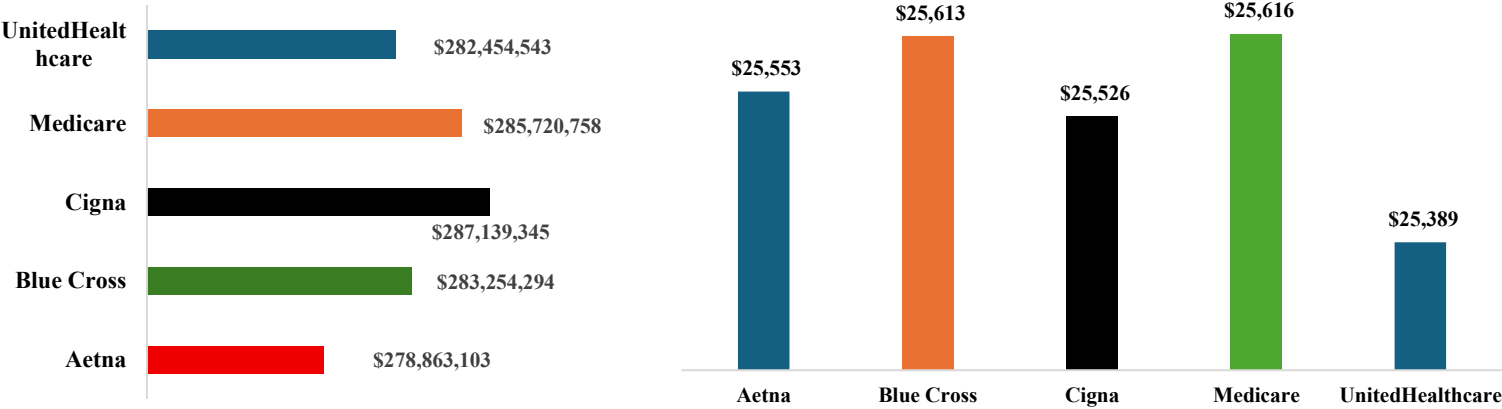


Figure 1. Q1 & Q2 — Total and average billing by insurance provider.

Q3 Do emergency admissions cost more than elective or urgent?

Domain: Financial & Revenue

A foundational assumption in hospital finance is that emergency admissions — due to unplanned care, diagnostic complexity, and resource intensity — should carry higher costs than planned (elective) visits. This analysis tests that assumption using both mean and median billing to account for outliers.

Admission Type	Mean Billing	Median Billing	Std Dev	Encounters
Elective	\$25,603	\$25,657	\$14,226	18,655
Urgent	\$25,518	\$25,550	\$14,241	18,576
Emergency	\$25,498	\$25,380	\$14,162	18,269

KEY FINDING: Contrary to expectations, elective admissions carry the highest average billing (\$25,603) while emergency admissions are the least expensive (\$25,498). The difference across all three types is only \$105 — not clinically or financially meaningful. This challenges the common assumption that emergency care is inherently more expensive.

RECOMMENDATION: Investigate whether case mix complexity differs by admission type. The flat billing pattern may indicate that DRG-based payment bundling is equalizing costs across acuity levels, or that documentation of emergency complexity is inconsistent.

Q4 Which medical conditions are associated with the highest billing amounts?

Domain: Financial & Revenue

Condition-level billing analysis helps identify high-cost patient populations for care management targeting, service line investment decisions, and population health program prioritization.

Medical Condition	Avg Billing	Total Revenue	Patients
Obesity	\$25,807	\$238.2M	9,231
Diabetes	\$25,639	\$238.5M	9,304
Asthma	\$25,636	\$235.5M	9,185
Hypertension	\$25,498	\$235.7M	9,245
Arthritis	\$25,498	\$237.3M	9,308
Cancer	\$25,163	\$232.2M	9,227

KEY FINDING: Obesity carries the highest average billing (\$25,807) and Cancer the lowest (\$25,163). The spread of \$644 across all six conditions suggests billing is primarily volume-driven rather than condition-specific — a signal that case mix differentiation may be limited in this dataset.

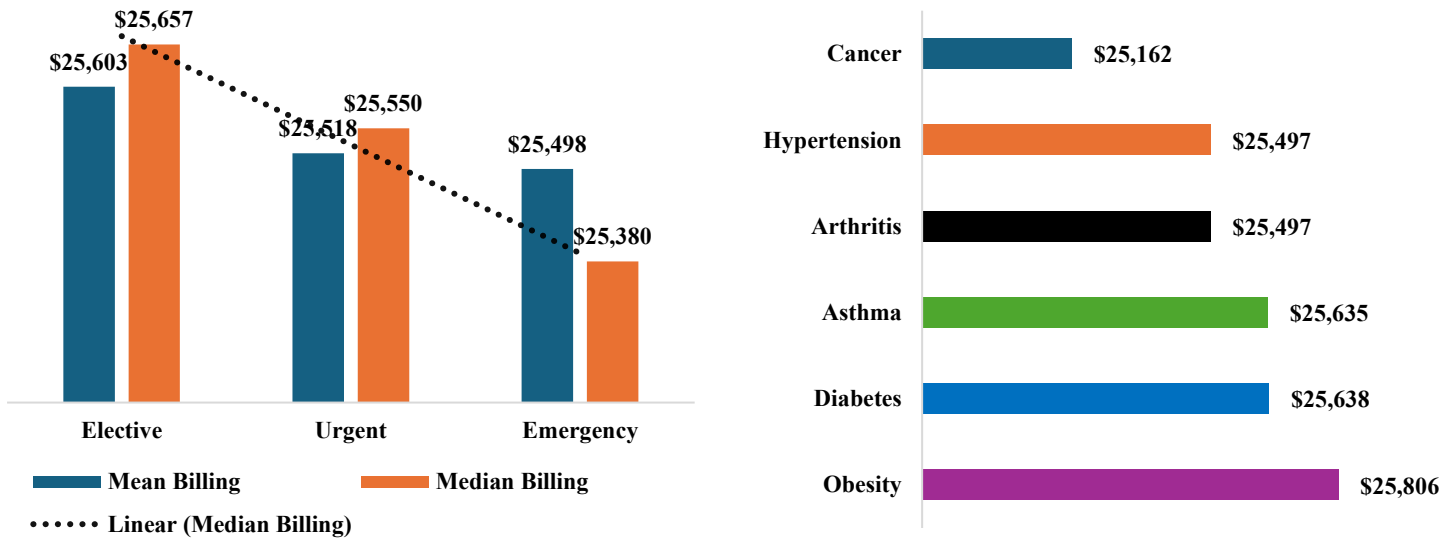


Figure 2. Q3 & Q4 — Billing by admission type and medical condition.

SECTION 2: UTILIZATION & OPERATIONS ANALYSIS

Q5 What is the average length of stay by admission type?

Domain: Utilization & Revenue

Length of stay (LOS) is one of the most operationally important metrics in hospital management. It drives bed capacity planning, staffing ratios, and discharge readiness protocols. Differences in LOS by admission type reveal whether acuity-based care pathways are functioning as designed.

Admission Type	Mean LOS (days)	Median LOS (days)	Std Dev
Emergency	15.6	16.0	8.67
Elective	15.5	15.0	8.65
Urgent	15.4	15.0	8.66

KEY FINDING: All three admission types show nearly identical LOS: Emergency 15.6 days, Elective 15.5 days, Urgent 15.4 days. The 0.2-day spread is not clinically significant. Acuity level does not predict discharge timing in this dataset — a major operational finding.

RECOMMENDATION: Review discharge planning and care pathway protocols. Identical LOS across acuity levels suggests that differentiated post-acute discharge planning is not occurring. Emergency patients — who may have more complex social determinants — may be held longer than clinically necessary, or elective patients are being managed with the same intensity as emergencies.

Q6 Are older patients associated with longer hospital stays or higher costs?

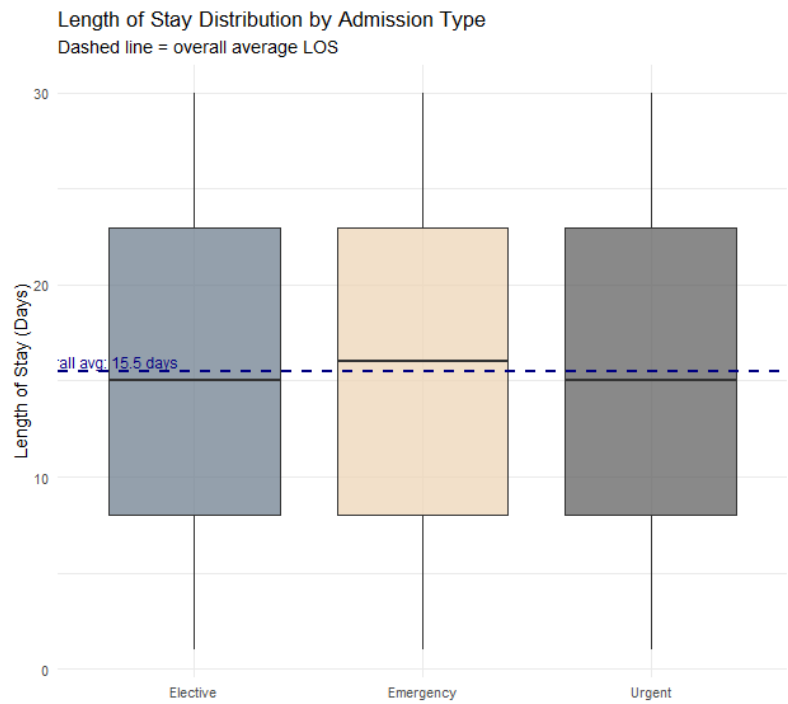
Domain: Utilization & Revenue

Age is a well-established predictor of health complexity and resource utilization in many healthcare settings. Older patients typically present with more comorbidities, require more diagnostics, and have more complex discharge needs. This analysis tests that relationship using both correlation coefficients and age-group level comparisons.

Age Group	Avg LOS (days)	Avg Billing (\$)	Encounters
≤34	15.4	\$25,597	12,486
35–49	15.4	\$25,560	11,834
50–64	15.6	\$25,621	14,108
65+	15.6	\$25,422	17,072

Statistical tests: Age vs. LOS: $r = 0.008$, $p = 0.053$ (not significant). Age vs. Billing: $r = -0.004$, $p = 0.368$ (not significant).

KEY FINDING: Age has no meaningful relationship with either LOS or billing in this dataset. The correlation coefficients are near zero and not statistically significant. Patients aged 65+ do not cost more or stay longer than younger patients — a finding that runs counter to general population health patterns and may reflect the simulated nature of the data.



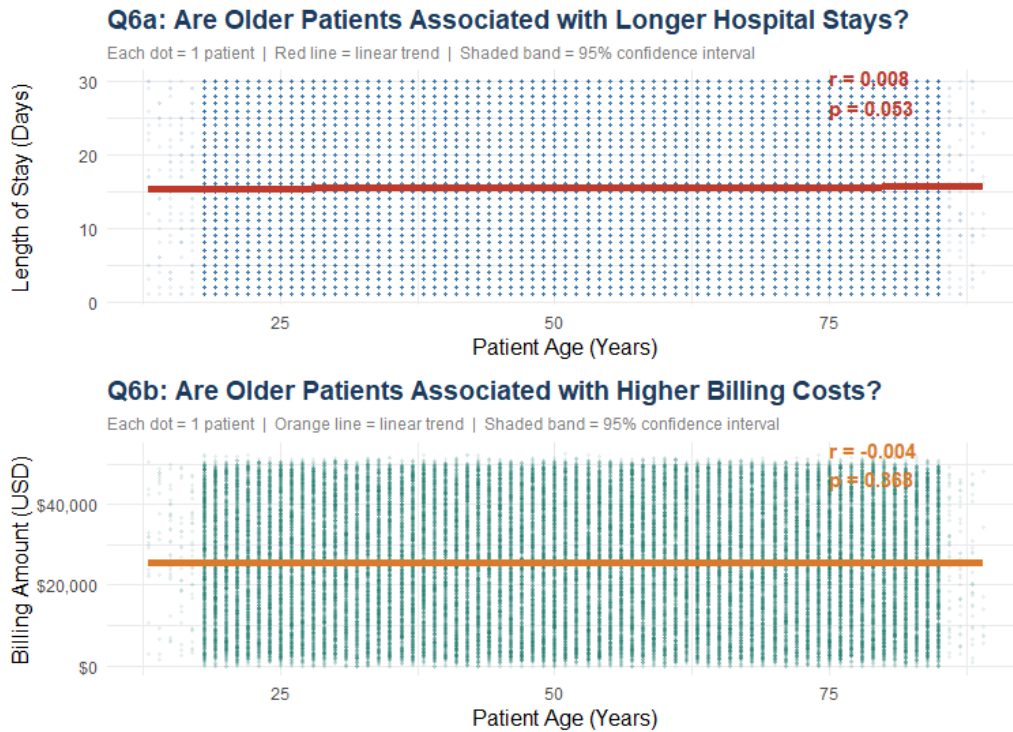


Figure 3. Q5 & Q6 — LOS by admission type and utilization by age group.

Q7 Which hospitals handle the highest volume of emergency admissions?

Domain: Utilization & Revenue

Hospital-level emergency volume analysis supports network capacity planning, facility performance benchmarking, and emergency department resource allocation. In this dataset, the hospital field contains ~39,876 unique hospital names — reflecting a highly fragmented provider landscape.

Rank	Hospital	Emergency Admissions
1	LLC Smith	19
2	Smith Ltd	14
3	Smith PLC	12
4	Johnson PLC	12
5	Inc Jones	12
6	Wilson Group	12
7	Ltd Smith	11
8	Johnson LLC	11
9	Inc Johnson	11
10	Miller PLC	11

KEY FINDING: With ~39,876 unique hospital identifiers, most hospitals appear only 1–3 times in the dataset. The top hospital (LLC Smith) handles only 19 emergency encounters. This extreme fragmentation prevents meaningful facility-level benchmarking and suggests the hospital field may contain generated names rather than a real-world network.

RECOMMENDATION: For real-world application of this analysis, hospital names should be standardized and mapped to NPI numbers or hospital systems. Network consolidation into health system groupings (e.g., HCA, Ascension, CommonSpirit) would enable meaningful volume and quality benchmarking.

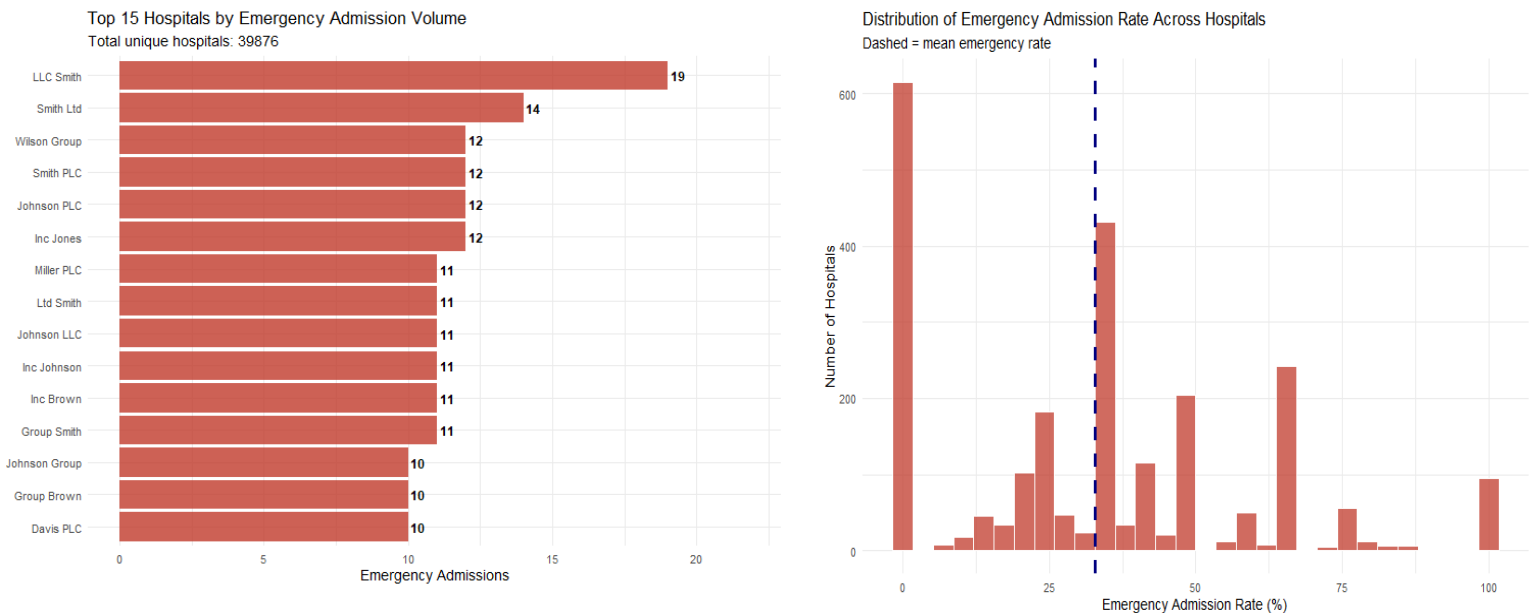


Figure 4. Q7 — Top hospitals by emergency volume and distribution of emergency rates.

SECTION 3: QUALITY & OUTCOMES ANALYSIS

Q8 Are abnormal test results associated with higher billing or longer stays?

Domain: *Quality & Revenue*

In clinical quality analytics, abnormal test results are hypothesized to signal greater diagnostic complexity, more interventions, and higher resource use. This analysis tests whether patients with abnormal results incur higher costs or longer stays than those with normal or inconclusive results, using one-way ANOVA to test for statistically significant differences.

Test Result	Avg Billing	Avg LOS (days)	Encounters
Abnormal	\$25,539	15.5	18,627
Inconclusive	\$25,625	15.5	18,356
Normal	\$25,458	15.5	18,517

Statistical tests: Billing ANOVA — $F = 0.64$, $p = 0.528$ (not significant). LOS ANOVA — $F = 0.38$, $p = 0.684$ (not significant at $p < 0.05$).

KEY FINDING: No statistically significant relationship exists between test result and billing ($p = 0.528$) or LOS ($p = 0.684$). Patients with abnormal results cost no more and stay no longer than those with normal results. This uniformity is analytically unusual and likely reflects the synthetic nature of the dataset — in real hospital data, abnormal results typically drive additional workup and cost.

Q9 Does admission type influence test outcomes?

Domain: *Quality & Revenue*

If admission type reflects clinical acuity, emergency patients would be expected to have a higher proportion of abnormal test results. This analysis uses a chi-square test of independence to examine whether the distribution of test outcomes differs significantly across admission types.

Admission Type	% Normal	% Abnormal	% Inconclusive
Elective	33.5%	33.8%	32.8%
Emergency	33.4%	33.3%	33.2%
Urgent	33.2%	33.6%	33.2%

Chi-square test: $\chi^2 = 1.32$, $df = 4$, $p = 0.858$ — not significant.

KEY FINDING: Admission type has no significant effect on test outcome distribution ($p = 0.858$). All three admission types show a near-identical ~33% split across Normal, Abnormal, and Inconclusive — effectively a uniform random distribution. This is strong evidence that test outcomes in this dataset are not clinically differentiated by admission acuity.

Figure 5. Q8 & Q9 — Test results by billing, LOS, and admission type.

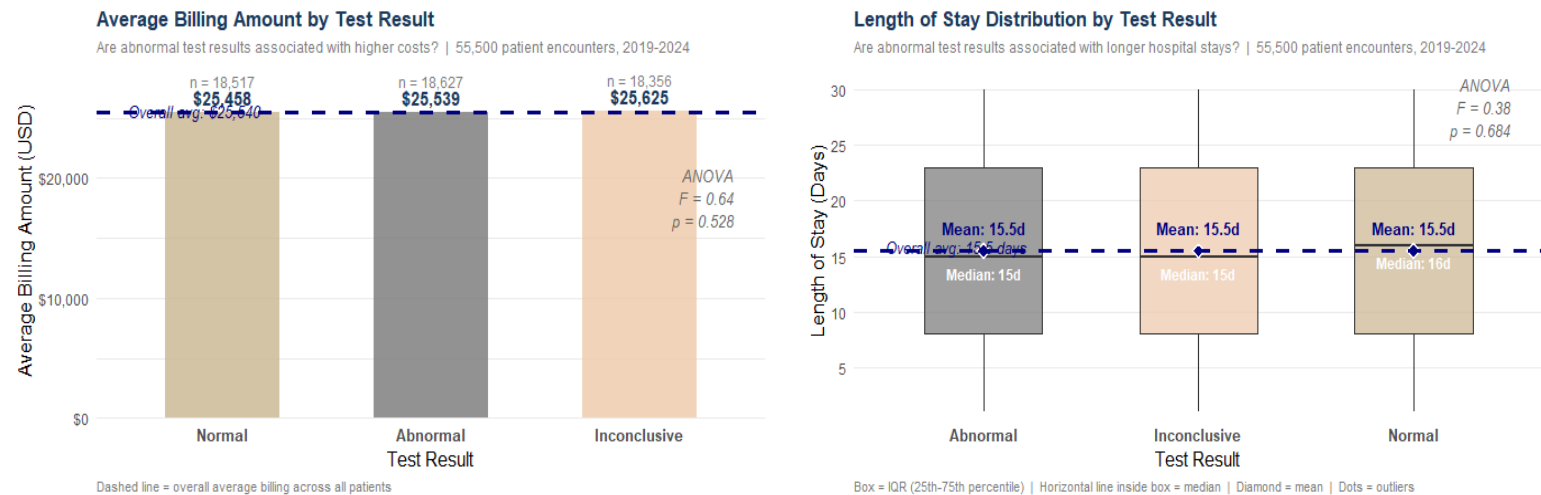


Figure 4. Q8a & Q8b — Test results by billing and LOS.

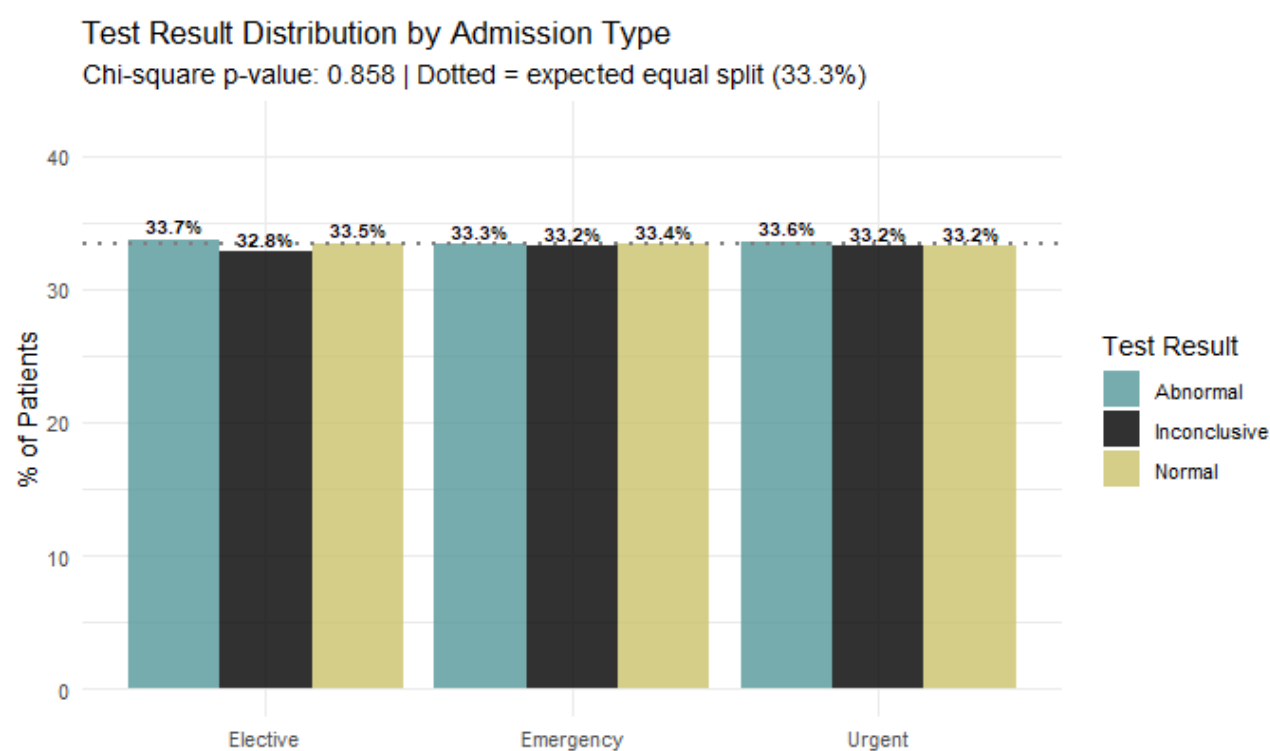


Figure 5. Q9 — Test results by admission type.

Q10 Are certain conditions more likely to result in emergency admissions?

Domain: *Quality & Revenue*

This question tests whether specific medical conditions drive emergency utilization — a key input for population health management programs that target high-risk patient groups for preventive interventions and care coordination to reduce avoidable emergency encounters.

Medical Condition	Emergency Rate	Elective Rate	Urgent Rate
Obesity	33.9%	33.6%	32.5%
Arthritis	33.4%	33.5%	33.1%
Asthma	32.7%	33.8%	33.5%
Cancer	32.7%	33.5%	33.8%
Hypertension	32.5%	33.6%	33.9%
Diabetes	32.4%	33.6%	34.0%

Chi-square test: $\chi^2 = 19.34$, $df = 10$, $p = 0.036$ — statistically significant.

KEY FINDING: The relationship between condition and admission type is statistically significant ($p = 0.036$), but practically small. Obesity has the highest emergency admission rate (33.9%) while

Diabetes has the lowest (32.4%) — a gap of only 1.5 percentage points. Statistical significance is driven by the large sample size (55,500), not a clinically meaningful difference.

RECOMMENDATION: *Despite the small effect size, obesity's position as both the highest-billing condition and the highest emergency admission rate makes it a dual-priority target for population health investment. Chronic disease management programs for obesity could yield both quality and financial returns.*

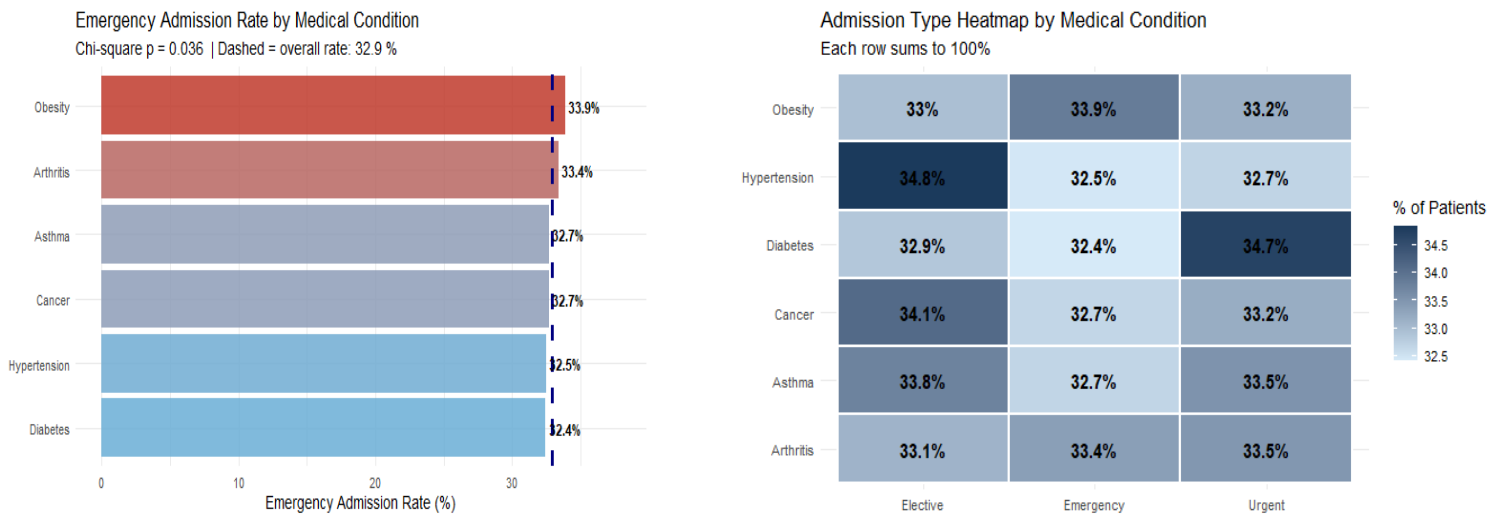


Figure 6. Q10 — Emergency admission rates by condition and condition × admission type heatmap.

Complete Findings Summary

#	Question	Key Finding	Significance
Q1	Total billing by insurer	Cigna leads (\$287.2M); all payers within \$8.3M range	Balanced payer mix
Q2	Avg billing by insurer	Medicare highest (\$25,617); spread of only \$227	Effectively uniform
Q3	Billing by admission type	Elective highest (\$25,603); Emergency lowest (\$25,498)	Difference = \$105 only
Q4	Billing by condition	Obesity highest (\$25,807); Cancer lowest (\$25,163)	Spread = \$644
Q5	LOS by admission type	All types $\approx$ 15.5 days; <0.2 day difference	No acuity differentiation
Q6	Age and utilization	$r \approx 0$ for both LOS and billing; age is not a predictor	Not significant
Q7	Hospital emergency volume	$\sim 40K$ unique hospitals; max 19 emergencies per facility	Fragmented network
Q8	Abnormal tests vs cost/LOS	No significant difference; $p = 0.528$ (billing)	$F = 0.64$, $p = 0.528$
Q9	Admission type vs test results	$\sim 33\%$ split in all categories for all admission types	$\chi^2 = 1.32$, $p = 0.858$
Q10	Condition vs emergency rate	Obesity highest (33.9%); 1.5% gap across conditions	$\chi^2 = 19.34$, $p = 0.036$

Strategic Recommendations for Decision-Makers

► Discharge Planning Protocol Review

Identical LOS across all three admission types is a critical operational finding. Acuity-differentiated discharge pathways should be reviewed to ensure emergency patients are not being managed with the same intensity and timeline as elective cases.

► Pricing Model Audit

Near-identical billing across all conditions, payers, and admission types suggests pricing may not be capturing clinical complexity differentials. A case mix and DRG audit is warranted to ensure appropriate risk and resource capture.

► Obesity as Dual-Priority Target

Obesity represents both the highest average billing (\$25,807) and the highest emergency admission rate (33.9%) among all conditions. Preventive care management programs targeting obesity would likely generate both quality improvement and financial returns.

► Provider Network Standardization

The ~40,000 hospital identifiers prevent meaningful facility benchmarking. Mapping hospitals to NPI numbers and health system affiliations would unlock network-level performance analytics, quality comparisons, and contract negotiation insights.

► Clinical Data Validation

The uniform distribution of test results (~33% each) and age-independent utilization patterns are analytically atypical and suggest a need for data quality review. In real-world healthcare analytics, abnormal results, older age, and emergency acuity should be stronger predictors.

APPENDIX: DATASET DESCRIPTION & METHODS

Data Source

Healthcare_Analytics.xlsx — hospital patient encounter records (simulated EHR + billing data).

Dataset Structure

Variable	Type	Description
Name / Name2	String	Patient identifier (raw and cleaned)
Age	Integer	Patient age at admission (13–89)
Gender	Categorical	Male / Female
Blood Type	Categorical	ABO + Rh blood group
Medical Condition	Categorical	Primary diagnosis (6 conditions)
Date of Admission	Date	Admission date (2019–2024)
Discharge Date	Date	Discharge date
LOS (derived)	Integer	Discharge Date minus Admission Date (days)
Doctor	String	Treating physician name
Hospital	String	Facility name (~39,876 unique)
Insurance Provider	Categorical	5 payers (Cigna, Medicare, UHC, Blue Cross, Aetna)
Billing Amount	Float	Total billed amount (USD); 1 negative value clipped to \$0
Admission Type	Categorical	Elective / Emergency / Urgent
Medication	Categorical	Primary medication prescribed
Test Results	Categorical	Normal / Abnormal / Inconclusive

Statistical Methods

- Descriptive statistics: mean, median, standard deviation, frequency counts
- Pearson correlation: for continuous variable relationships (Age vs. LOS, Age vs. Billing)
- One-way ANOVA: for comparing means across 3+ groups (Test Results vs. Billing/LOS)
- Chi-square test of independence: for categorical associations (Admission Type vs. Test Results; Condition vs. Admission Type)
- All analyses and visualizations conducted in Excel and R (tidyverse, readxl, rstatix, scales, ggthemes)

Prepared by: Oluwatobi A. Alawode, PhD · May 2026